IGCSE 0607 Practice Questions — E3.4

Finding a quadratic function using given information

Total Questions: 39 — **Total Marks:** 210 Marks

Table of Contents & Quick Navigation

- **E3.4** Finding a quadratic function using given information
  → **Paper 2 (Non-Calculator)** *(21 Qs)* · **Paper 4 (Calculator)** *(18 Qs)*

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1 A quadratic function has x-intercepts at $x = -3$ and $x = 4$.
The coefficient of x^2 is 1.

Write down the equation of this function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[2]**

2 A quadratic function has x-intercepts at $x = 4$ and $x = -2$.
The coefficient of x^2 is 1.

Find the equation of the quadratic function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

3 A quadratic function $f(x)$ has x-intercepts at $x = -4$ and $x = 3$.
The coefficient of x^2 is 1.

Find the equation of the function in the form $f(x) = x^2 + bx + c$.

$f(x) =$ ..... **[2]**

4 A quadratic function has x-intercepts at $x = -4$ and $x = 7$.
The coefficient of x^2 is 1.
Write down the equation of the function.
Give your answer in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

5 A quadratic function has the equation $y = x^2 + bx + c$.
The vertex of the graph of this function is $(4, -3)$.

Find the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

6 A quadratic function is given by

$y = x^2 + bx + c$

The vertex of the graph of this function is $(3, -4)$.

Find the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

7 A quadratic function has a minimum point at $(3, -4)$ and passes through the point $(5, 0)$.
The coefficient of x^2 is 1.

Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[3]**

8 The graph of a quadratic function, $y = x^2 + bx + c$, crosses the x-axis at $(-4, 0)$ and $(2, 0)$.

(a) Work out the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Find the coordinates of the vertex of the graph.

(..... , ...) **[2]**

9 The graph of a quadratic function $f(x) = x^2 + bx + c$ crosses the x-axis at $(-3, 0)$ and $(5, 0)$.

(a) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[3]**

(b) Write down the coordinates of the y-intercept of the graph of $y = f(x)$.

(..... , ...) **[1]**

10 A quadratic function $y = f(x)$ has its vertex at $(3, -4)$.
The coefficient of x^2 is 1.

(a) Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis.

..... **[2]**

11 A quadratic function is given by the equation $y = (x - h)^2 + k$.
The vertex of the graph of this function is $(3, -2)$.

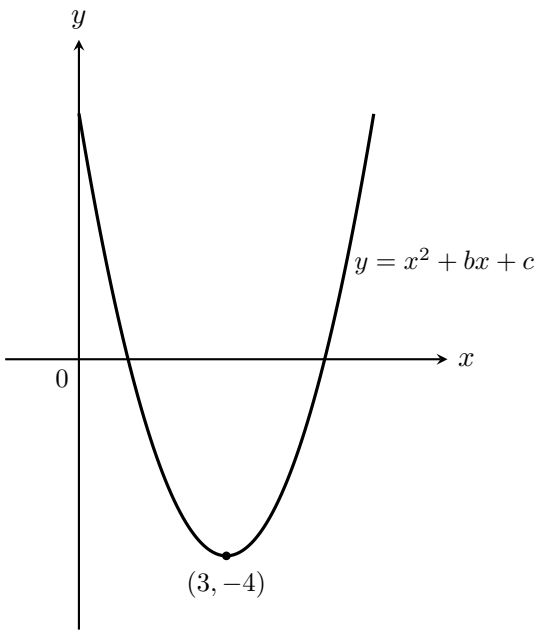
(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) The graph passes through the point $(5, p)$.
Work out the value of p.

$p =$ ..... **[2]**

12 The diagram shows a sketch of the quadratic curve $y = x^2 + bx + c$.
The vertex of the curve is at $(3, -4)$.



(a) Write down the equation of the curve in the form $y = (x - h)^2 + k$.

$y =$ ..... **[2]**

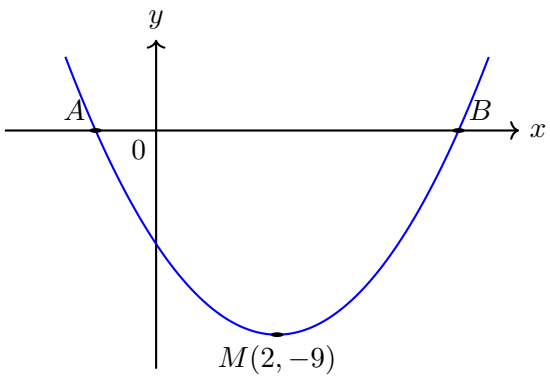
(b) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[2]**

(c) Write down the coordinates of the point where the curve crosses the y-axis.

(..... , ...) **[1]**

13 The diagram shows the graph of a quadratic function $y = x^2 + px + q$.
The graph has a minimum point (vertex) at $M(2, -9)$.



(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(b) The graph crosses the x-axis at the points A and B.
Find the coordinates of A and the coordinates of B.

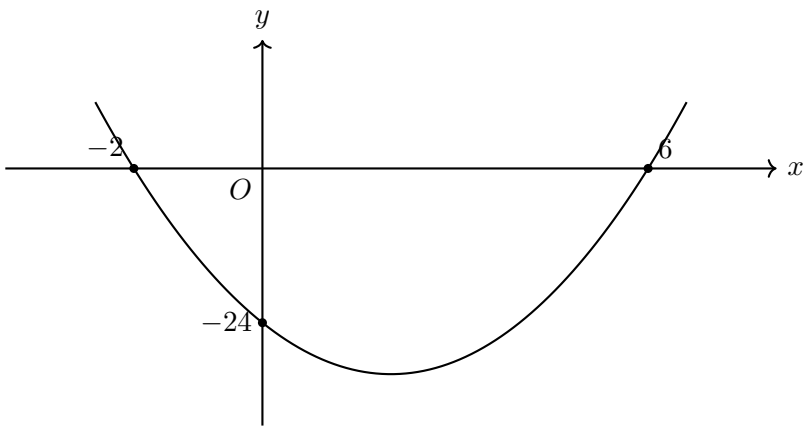
$A =$ ..... , $B =$ ..... **[3]**

14 The graph of a quadratic function $y = f(x)$ intersects the x-axis at $(-2, 0)$ and $(5, 0)$.
The graph intersects the y-axis at $(0, -20)$.

Work out the function $f(x)$, giving your answer in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

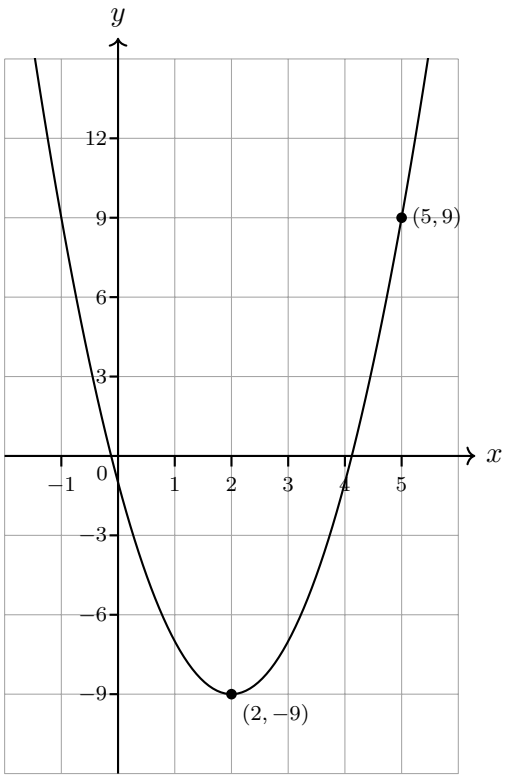
15 The diagram shows a sketch of the quadratic graph $y = f(x)$.
The graph crosses the x-axis at $(-2, 0)$ and $(6, 0)$.
It crosses the y-axis at $(0, -24)$.



Find the expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

16 The diagram shows the graph of a quadratic function $g(x)$.
The vertex of the parabola is at $(2, -9)$ and the curve passes through the point $(5, 9)$.



(a) Find the equation of the quadratic function in the form $g(x) = a(x - h)^2 + k$.

$g(x) =$ ..... **[3]**

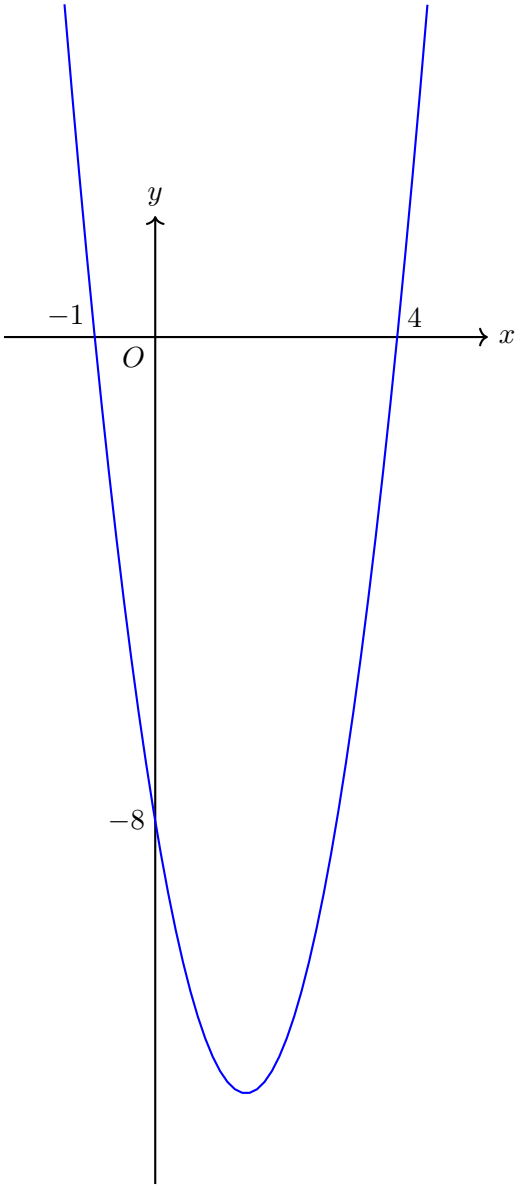
(b) Express $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[2]**

(c) Write down the equation of the line of symmetry of the graph of $y = g(x)$.

..... **[1]**

17 The diagram shows a sketch of the quadratic function $y = f(x)$.
The graph crosses the x-axis at $(-1, 0)$ and $(4, 0)$.
The graph crosses the y-axis at $(0, -8)$.



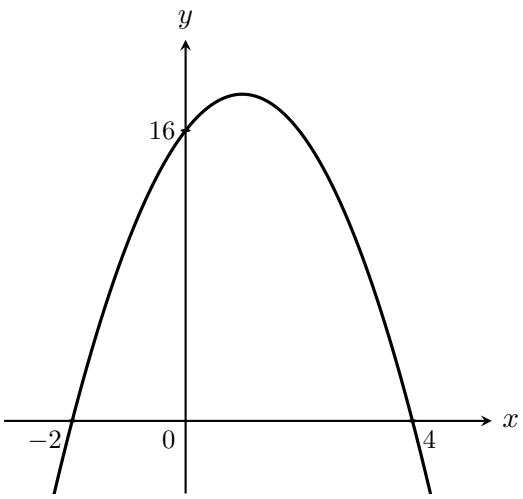
(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

(b) Find the coordinates of the minimum point of the graph.

(..... , ...) **[3]**

18 The diagram shows the graph of a quadratic function $y = ax^2 + bx + c$.
The curve crosses the x-axis at $(-2, 0)$ and $(4, 0)$, and crosses the y-axis at $(0, 16)$.



(a) Find the equation of the curve in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

(b) Find the coordinates of the turning point of the curve.

(..... , ...) **[2]**

(c) Write down the equation of the line of symmetry of the curve.

..... **[1]**

19 A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The graph of $y = f(x)$ has a maximum point at $(-1, 8)$ and passes through the point $(2, -10)$.

(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

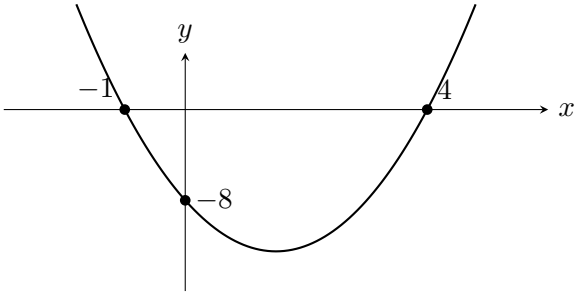
(b) Find the value of a.

$a =$ ..... **[3]**

(c) Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[3]**

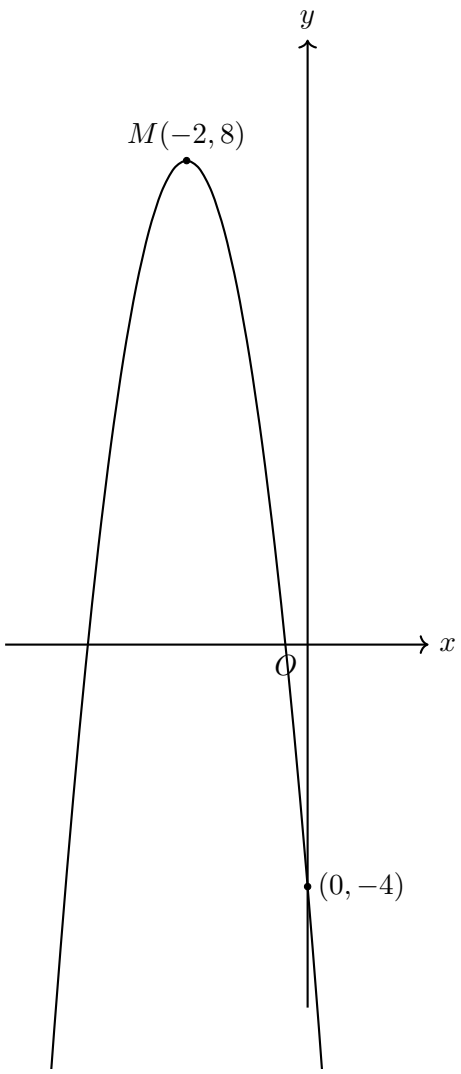
20 The diagram shows the graph of a quadratic function $y = f(x)$.
The graph crosses the x-axis at $x = -1$ and $x = 4$.
The graph crosses the y-axis at $(0, -8)$.



Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

21 The diagram shows a sketch of the curve $y = f(x)$ where $f(x)$ is a quadratic function.
The maximum point on the curve is $M(-2, 8)$ and the curve crosses the y-axis at $(0, -4)$.



(a) Find the equation of the curve.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Solve $f(x) = 0$ to find the exact x-coordinates of the points where the curve crosses the x-axis.

Give your answers in the form $\frac{p \pm q\sqrt{6}}{r}$, where p, q and r are integers.

$x =$ ..... **[3]**

22 A quadratic curve has its vertex at $(3, -5)$ and passes through the point $(1, 7)$.
Find the equation of the curve in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

23 The graph of a quadratic function has its vertex at $(-2, 5)$ and passes through the point $(1, -13)$. Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

24 The graph of a quadratic function has its vertex at $(2, 7)$ and passes through the point $(4, -5)$.

(a) Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

(b) Find the y-intercept of the graph.

..... **[1]**

25 A quadratic curve crosses the x-axis at $(-1, 0)$ and $(5, 0)$.
The curve also passes through the point $(2, -18)$.
Find the equation of the curve in the form $y = ax^2 + bx + c$.

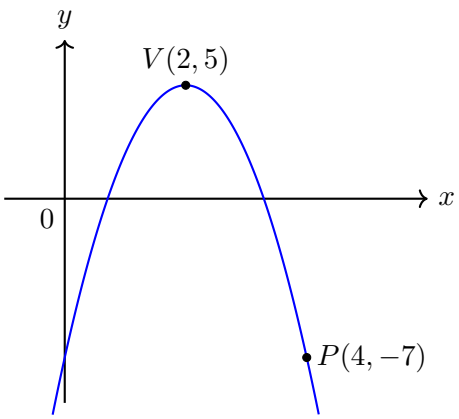
$y =$ ..... **[4]**

26 A quadratic curve has a vertex at the point $(2, -5)$ and passes through the point $(4, 7)$.

Find the equation of the curve.
Give your answer in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

27 The diagram shows a sketch of the graph of a quadratic function $y = a(x - h)^2 + k$.
The vertex of the graph is $V(2, 5)$.
The graph passes through the point $P(4, -7)$.



(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

(c) Find the coordinates of the y-intercept of the graph.

(..... , ...) **[2]**

28 The graph of a quadratic function $f(x)$ intersects the x-axis at $x = -2$ and $x = 5$.
The graph passes through the point $(3, 10)$.

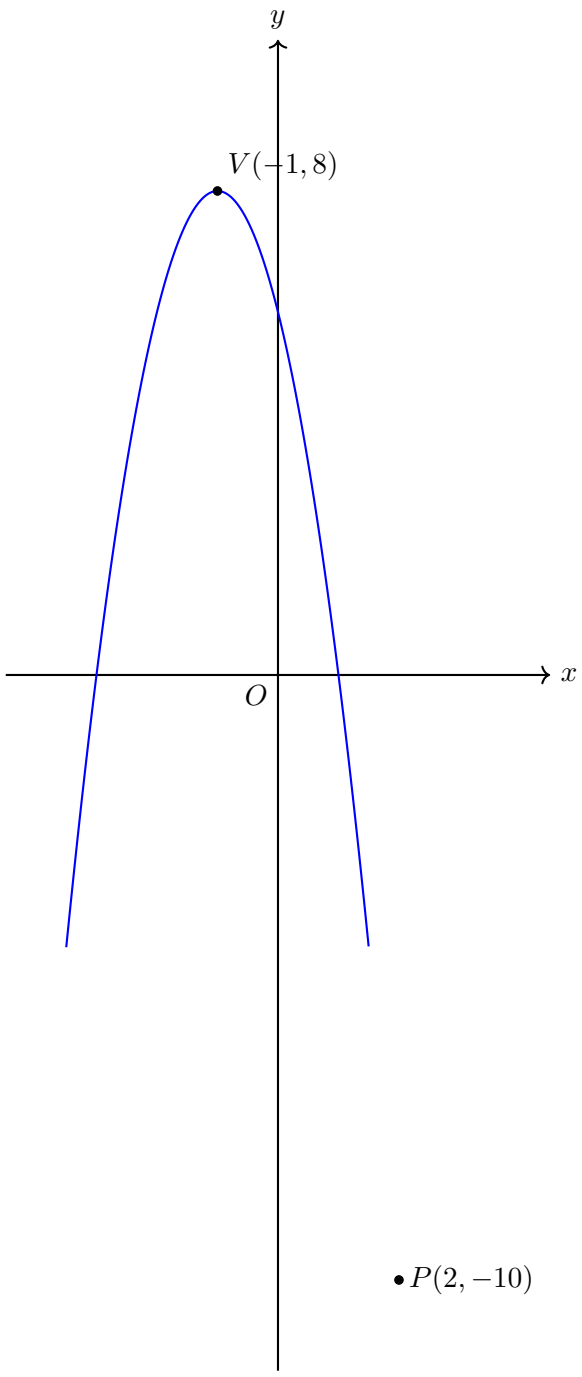
(a) Find an expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Find the coordinates of the vertex of the graph of $y = f(x)$.

(..... , ...) **[2]**

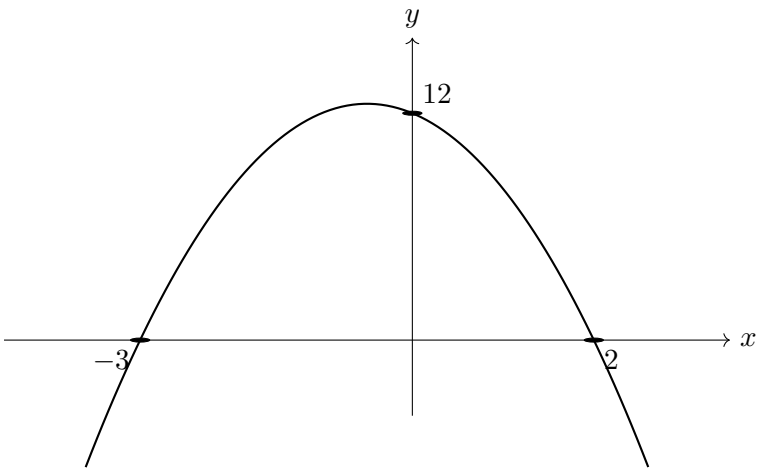
29 The diagram shows a sketch of the graph of $y = ax^2 + bx + c$.
The vertex of the graph is $V(-1, 8)$ and it passes through the point $P(2, -10)$.



Work out the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

30 The diagram shows the sketch of a quadratic function.
The curve crosses the x-axis at -3 and 2, and intersects the y-axis at 12.



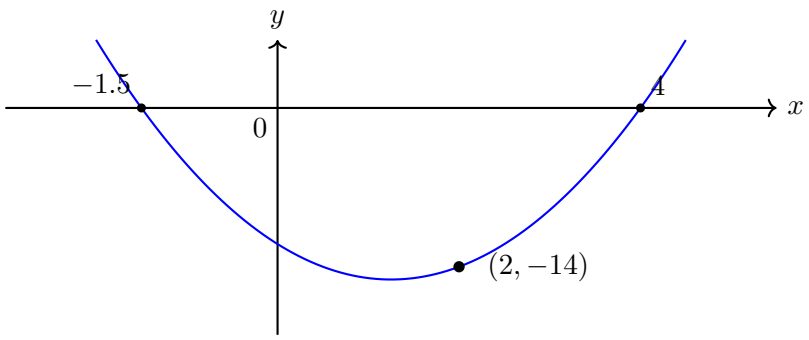
(a) Find the equation of the quadratic function.
Give your answer in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

(b) Calculate the coordinates of the vertex of this curve.

((..... , ...)) **[2]**

31 The diagram shows a sketch of the quadratic function $y = f(x)$.
The curve crosses the x-axis at $x = -1.5$ and $x = 4$.
It passes through the point $(2, -14)$.



(a) Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Find the exact coordinates of the turning point of the graph.

(..... , ...) **[3]**

32 A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The maximum point of the graph of $y = f(x)$ is $(-4, 20)$.
The graph intersects the x-axis at $x = -2$ and $x = p$.

(a) (i) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(ii) Find the value of a.

$a =$ ..... **[2]**

(iii) Find the value of p.

$p =$ ..... **[1]**

(iv) Express $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

33 A quadratic function $h(x) = ax^2 + bx + c$ has x-intercepts at $(-2, 0)$ and $(4, 0)$, and passes through the point $(1, 13.5)$.

(a) Find the values of a, b, and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[4]**

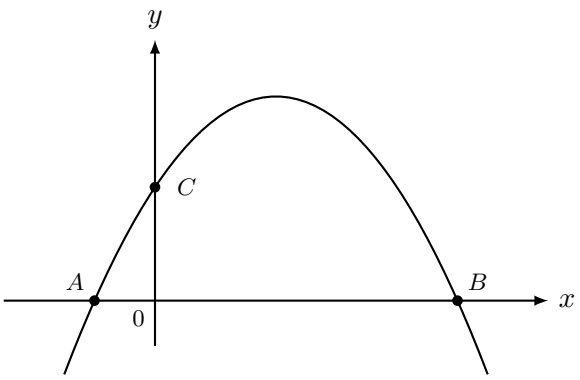
(b) Hence find the coordinates of the vertex of the graph of $y = h(x)$.

(..... , ...) **[2]**

(c) State the range of $h(x)$.

..... **[1]**

34 The graph of a quadratic function $y = f(x)$ intersects the x-axis at the points $A(-1, 0)$ and $B(5, 0)$. The graph intersects the y-axis at the point $C(0, 10)$.



(a) Find the equation of the function $f(x)$.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

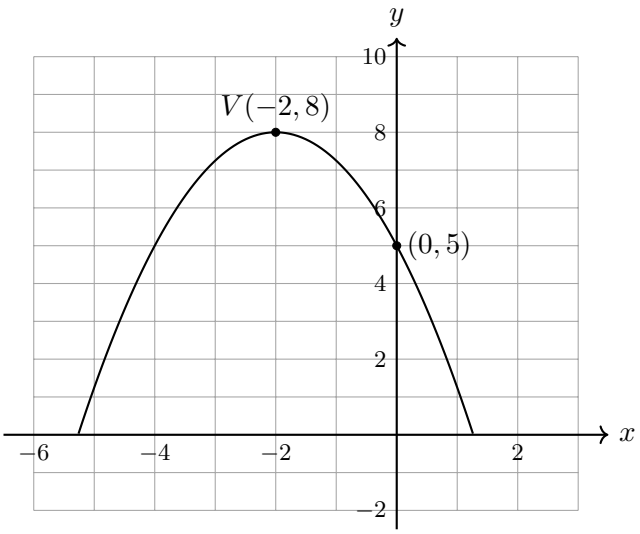
(b) Find the coordinates of the turning point (vertex) of the graph.

(..... , ...) **[2]**

(c) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find the equation of $g(x)$ in the form $g(x) = a(x - h)^2 + k$.

$g(x) =$ ..... **[2]**

35 The diagram shows the graph of a quadratic function $y = f(x)$.
The vertex of the graph is at $V(-2, 8)$ and it intersects the y-axis at $(0, 5)$.



(a) Find the equation of the function $f(x)$ in the form $f(x) = a(x - h)^2 + k$.

$f(x) =$ ..... **[3]**

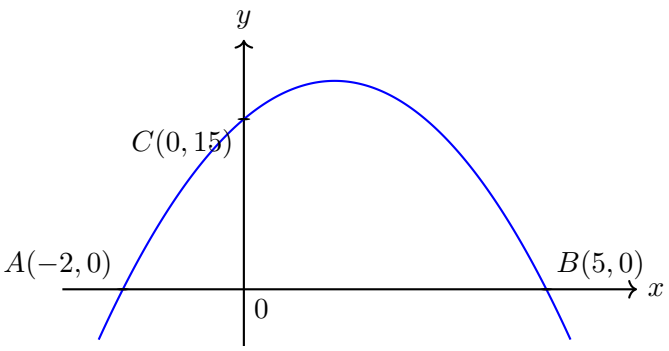
(b) Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

(c) Calculate the exact x-coordinates of the points where the graph intersects the x-axis.
Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[3]**

36 The diagram shows a sketch of the quadratic curve $y = ax^2 + bx + c$.
The curve crosses the x-axis at $A(-2, 0)$ and $B(5, 0)$, and crosses the y-axis at $C(0, 15)$.



(a) Find the values of a, b, and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

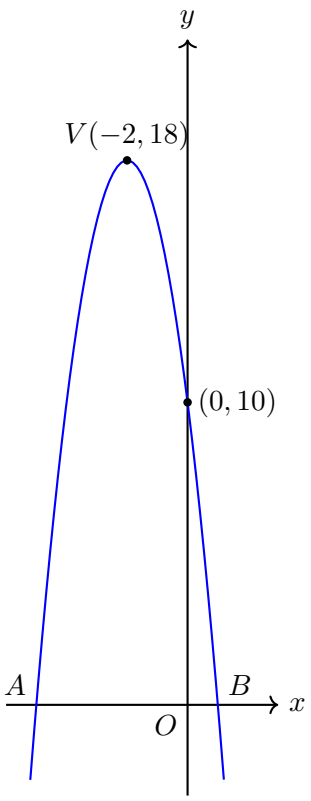
(b) Find the coordinates of the vertex of the curve.

(..... , ...) **[3]**

(c) The equation $ax^2 + bx + c = k$ has no real roots.
Write down the inequality for k.

..... **[1]**

37 The diagram shows a sketch of the graph of a quadratic function $y = f(x)$.
The maximum point of the graph is $V(-2, 18)$.
The graph crosses the y-axis at $(0, 10)$.



(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

(b) The graph crosses the x-axis at A and B.
Find the length of the line segment AB.

$AB =$ ..... **[3]**

(c) The line $y = -2x + k$ is a tangent to the curve $y = f(x)$.

Find the value of k.

$k =$ ..... **[4]**

38 A quadratic function $f(x) = a(x-h)^2 + k$ has its vertex at $(4, -5)$ and passes through the point $(1, 8.5)$.

(a) Find the values of a, h, and k.

$a =$ ..... , $h =$ ..... , $k =$ ..... **[3]**

(b) Solve the equation $f(x) = 0$, giving your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

(c) The line $y = 2x - 7$ intersects the curve $y = f(x)$ at two points, P and Q.
Find the x-coordinates of P and Q, giving your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

(d) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -3 \\ 9 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in the form $p(x-q)^2 + r$.

$g(x) =$ ..... **[2]**

39 The quadratic function $f(x) = ax^2 + bx + c$ has a minimum turning point at $(1, -8)$.
The graph of $y = f(x)$ crosses the x-axis at $x = -1$ and $x = 3$.

(a) Show that $f(x) = 2x^2 - 4x - 6$.

[3]

(b) Solve the equation.

$f(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

(c) A straight line has equation $y = mx - 8$.
The line is a tangent to the graph of $y = f(x)$.
Find the possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

(d) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ k \end{pmatrix}$.
The new graph has no x-intercepts.
Write down the inequality for k.

..... **[2]**